

SENSE

QR Tech Solutions

Delivery Report: Platform Redesign, Cloud, Zones, Roadmap

7th of September 2026 · Completed work and future plans

1. Report information

Field	Details
Prepared by	Amdjed Belgacem
Product	QR Tech Solutions Sense
Period	7 September 2026
Report type	Delivery report + future roadmap
For	Engineering / leadership review
Focus	Completed plans, platform redesign, competitive next steps

2. Purpose of this report

This report covers the Sense plans delivered on the 7th of dSeptember, including the platform redesign I pushed through the product UI. It also sets out future plans, especially in perception and classification, where Sense needs to compete with tools operators already compare us against.

I am not writing this as a claim that the product is finished. I am writing it as a clear record of what shipped, what still needs closing, and where the next advantage can come from.

3. Platform redesign

Beyond backend reliability work, I redesigned the platform surface so Sense feels like a product workspace rather than a pile of engineering pages.

- Dark product shell aligned with the 3D scene
- Clear separation between My Uploads and Other Sources
- Large point cloud scene as the center of the product
- Stronger navigation across scene, uploads, zones, analytics, and live sources
- Inspection first defaults: cloud visible, tracks optional
- Honest status states for processing, failed decode, empty sources, and corrections

Why this matters: perception software loses trust fast when the interface feels temporary. The redesign is part of making Sense presentable for internal review and partner conversations.

4. Plan 1: Upload reliability, Hesai path, live UDP

4.1 Completed successfully

- **Upload vanishing fixed:** My Uploads uses origin filtering so live bridge traffic no longer pushes user jobs out of the rail. Card selection stays sticky across polls.
- **Hesai JT class usable path:** JT128 style captures decode into visible geometry. Unsupported families fail cleanly and honestly.
- **Live unicast UDP:** Opt in desk sensor path exists. Multicast remains a documented gap.
- **Fusion:** Left as groundwork only. No fake multi lidar production claim.

5. Plan 2: Cloud display quality

5.1 Completed successfully

- **Denser JT128 clouds:** second return merge recovered the missing half of many packets.
- **False people reduced:** footprint and point gates cut wall and furniture spam. Tracks off by default for inspection.
- **Correction CSV path:** hardware angle style config can be attached and surfaced in the UI.
- **500k default budget:** with quality toggle for performance, default, and max.

This closed a major product gap against PandarView style inspection expectations. Sense still needs progressive loading polish so dense quality does not feel slow on first open.

6. Plan 3: Mock live data cleanup

Synthetic demo lidar is no longer presented like a real sensor. Demo sources are explicitly labeled. Normal product start stays clean.

7. Plan 4: Zones perfection

7.1 Phases A to D completed

- **Phase A:** validation, overlap independent counting, safe registry writes, no job crash on one bad zone.
- **Phase B:** rename, color, enable or disable, delete confirm, draft restore, shared selection highlight.
- **Phase C:** per source scope, overlay filtering, honest camera vs lidar frame labels.
- **Phase D:** stream keyed lidar counters, hub snapshot fix, source scoped events and analytics, camera ground safety.

7.2 Phase E closeout

Phase E finishes event truth and visual feedback:

- Record real ZoneCounter events with track ids into history
- CSV and throughput from that same series
- Heatmap bounds from cloud bounds when available
- Enter and exit feedback in the active point cloud view

When Phase E lands, zones should hold up under the original acceptance checks for draw, save, reload, enter, exit, overlaps, and source isolation.

8. Status snapshot

Item	Status	Notes
Platform redesign	DONE	Product shell and scene first workspace
Upload vanishing under load	DONE	Origin filter + sticky selection
Hesai JT class usable path	DONE	Honest unsupported failures
Live unicast UDP	DONE	Opt in
Dense cloud decode and budget	DONE	500k default + correction path
Tracks off by default	DONE	Inspection first
Mock live lidar labeled	DONE	Demo badge
Zones A to D	DONE	Correctness, editor, scope, multi source
Zones E	IN PROGRESS	Event truth and ripples
Progressive cloud loading	NEXT	Fast preview then refine
Premium UI polish pass	NEXT	Competitive visual quality
Advanced perception and classification	PLANNED	See roadmap section
Multicast live	NOT YET	Documented gap
Full multi lidar fusion	NOT YET	Deferred on purpose

9. Future plans

9.1 Near term product plans

- Finish Zones Phase E and lock acceptance checks
- Progressive point loading so dense clouds feel fast
- Premium UI polish across scene, uploads, zones, analytics, and live panels
- Meet with Mr Laith and Mr Anis to walk through QR Sense in action, align on how it works, and define the next implementation steps together
- Widen lidar brand coverage beyond the current working families
- Stronger packaging so someone outside the repo can run a clean demo

9.2 Competitive perception and classification plans

Many lidar and perception tools are strong in one slice: vendor viewers, pure visualization, or narrow analytics. Sense should compete by combining multi brand intake, operator UI, and perception that is useful in real sites.

- **Multi class 3D perception, not people only:** classify and track people, vehicles, clusters, and generic obstacles. Many counting tools stop at people. Site operations need a wider object language.
- **Semantic cloud coloring operators can trust:** true class coloring when available, with honest fallback when class data is missing. Competitors often either hide uncertainty or overclaim classification quality.
- **Scene understanding for operations:** zone occupancy, flow, loitering, queueing, and blocked path style events, not only enter and exit totals.
- **Cross sensor association without fake fusion claims:** camera and lidar evidence linked where calibration allows, while staying explicit about what is measured vs estimated.
- **Brand agnostic perception layer:** same zones, tracks, and classification outputs across Ouster, Velodyne, Hesai, and future vendors. Vendor tools usually stop at their own hardware.
- **Operator grade explainability:** why a track was created, why a zone fired, confidence and source of the decision. This is weak in a lot of closed perception stacks.
- **Dense scene robustness:** better behavior in clutter, furniture heavy rooms, and partial occlusions where classical detectors still confuse objects for people.
- **Online quality monitoring:** packet loss, correction missing, calibration drift hints, and decode health visible in the product. Many tools show a pretty cloud and hide the reliability story.

9.3 Capabilities competitors often have that Sense should match

- Fast dense inspection playback with clean timeline controls
- Measurement tools that feel precise in the cloud
- Easy correction and calibration import from vendor ecosystems
- Export paths for reports, clips, and selected cloud subsets
- Stable live bring up for real sensors, including harder network modes later

9.4 Capabilities Sense can push harder than typical tools

- One workspace for uploads, live sources, zones, counts, and analytics
- Software first multi brand direction instead of one vendor cage
- Browser based operational scene that does not force a heavy desktop only workflow
- Honest product language around support limits, corrections, and confidence
- Faster iteration toward site ready perception instead of research only demos

10. Why this roadmap can compete

The market already has vendor viewers, robotics visualizers, and narrow people counting products. The gap is a software layer that feels good to operate, accepts more than one lidar brand, shows real geometry, and turns perception into decisions sites can use.

If Sense keeps compounding on redesign quality, cloud trust, zones reliability, and broader classification, it can stand as more than a prototype. It can stand as the operational layer around lidar, not just another viewer.

11. Closing

On 7th September I delivered the reliability and product shape work Sense needed next: platform redesign, stable uploads, denser Hesai path clouds, live unicast UDP, mock data honesty, and zones phases A to D.

The next competitive edge is not only more buttons. It is better perception and classification, clearer operator trust, and a premium interface that makes those capabilities usable.

Immediate closeouts are Zones Phase E, progressive loading, and the working session with Mr Laith and Mr Anis. After that, the roadmap should push multi class perception and brand agnostic operational features hard.